

SOA Survey

Personalized Federated Learning for Cross-Market Portfolio Allocation: Who Benefits Under Heterogeneity?

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Abstract

Portfolio allocation models benefit from collaboration across institutions, but privacy regulations prevent sharing of raw trading data. Federated Learning (FL) enables collaborative model training without data sharing, yet standard FL aggregates all clients uniformly, a design that works when client data distributions are similar but degrades when they are structurally different. Financial markets are among the most heterogeneous datasets that exist: the KSE-100, S&P 500, FTSE-100, and Nikkei 225 differ in size, volatility, currency, and macro drivers. This survey reviews 11 papers spanning four challenge areas: the foundations of FL and the FedAvg algorithm, heterogeneity-robust FL (FedProx), personalised FL (Per-FedAvg), and FL applied to financial time series and portfolio optimisation. The gap we identify is that existing cross-market FL papers focus on engineering better aggregation mechanisms and report only aggregate performance. In the papers reviewed here, we did not find a client-level fairness or per-market benefit analysis for cross-market federated portfolio learning, nor a study of whether lightweight personalisation corrects such imbalances. We propose an empirical study of client-level asymmetry under cross-market federated portfolio learning, using publicly available data from four markets and six baselines evaluated per client.

1. Problem Definition

Portfolio allocation is the task of deciding, at each time step, what fraction of available capital to assign to each available asset so as to maximise risk-adjusted return over time. Unlike a simple buy-and-hold strategy, a portfolio allocation model must account for changing asset correlations, varying volatility regimes, transaction costs from rebalancing, and the possibility of capital loss. From a deep learning standpoint the task is a temporal sequence problem:

the model receives a window of recent return observations and must output a weight vector over the asset universe.

The motivation for collaborative learning across institutions is clear: any single brokerage or fund manager observes only a fragment of global market dynamics. A London-based manager sees FTSE 100 behaviour but not the co-movements of Japanese exporters or Pakistani blue-chips. A shared model trained across many institutions would capture broader patterns. However, raw financial data is commercially sensitive, legally regulated, and competitively guarded. No institution will upload its order book to a central server. Federated Learning directly addresses this constraint by keeping raw data local and sharing only model weight updates.

The deep learning challenge that motivates this survey arises at the intersection of FL and financial heterogeneity. Standard FL assumes that clients are *different in quantity* but similar in kind: each hospital has different numbers of MRI scans, but all scans come from similar patients. Financial markets break this assumption. The KSE-100 is a small, high-volatility, rupee-denominated emerging market. The S&P 500 is a large, liquid, dollar-denominated developed market. These two clients are not merely different in size; they are *structurally different*. Averaging their weight updates via FedAvg produces a global model that may be biased toward the larger, more data-rich market, potentially leaving smaller clients worse off than if they had trained locally. Whether this asymmetry exists, how large it is, and whether personalisation can correct it are the questions that drive this project.

2. Literature Review

2.1. Federated Averaging: The Baseline Protocol

The canonical starting point for all FL work is the FedAvg algorithm introduced by McMahan et al. [1]. FedAvg operates in rounds: the server broadcasts a global model to a fraction of clients; each client runs several local SGD steps on its own data; clients return updated weights; the server averages them by data volume. McMahan et al. showed that this scheme can train deep networks with 10–100 $\times$ fewer communication rounds than naive gradient sharing, and demonstrated it on MNIST, CIFAR-10, and a large-scale next-word-prediction task. The key insight is that multiple local update steps significantly reduce the number of communication rounds needed.

What FedAvg does not address is heterogeneity. The paper explicitly notes that results on non-IID partitions are worse than IID, and flags this as an open problem. This limitation is what motivated the entire sub-field of non-IID FL. For our project, FedAvg is the primary baseline we expect standard cross-market federation to resemble.

2.2. The Non-IID Heterogeneity Problem

The non-IID challenge in FL occurs when client data distributions are statistically different, causing local gradient updates to point in conflicting directions. The server average then represents no single client well, a phenomenon called *client drift*. Li et al. [2] formalised this in their FedProx paper, which adds a proximal term to each client’s local objective that penalises deviation from the global model. FedProx subsumes FedAvg as a special case (proximal coefficient set to zero) and provides convergence guarantees under heterogeneous data. Empirically, FedProx improves over FedAvg on synthetic and real non-IID datasets, though the gains depend on careful tuning of the proximal coefficient.

A comprehensive taxonomy of non-IID settings is provided by the survey of Jimenez G. et al. [5], which covers heterogeneity types including feature distribution shift, label distribution skew, quantity imbalance, and spatiotemporal skew. Critically for our project, the survey notes that *quantity imbalance* (where some clients have far more data than others) interacts de-

structively with other skew types in FedAvg, because large-client gradients dominate the average. Financial markets exhibit several of these heterogeneity types. Feature distributions differ across markets: return distributions and volatility regimes vary significantly between, say, the S&P 500 and the KSE-100. Quantity imbalance is also present, since the S&P 500 has decades of deep historical data while the KSE-100 has a shorter and sparser record. Spatiotemporal skew applies too, as markets operate in different time zones, respond to different macro drivers, and evolve at different rates. Label skew in the classification sense is less relevant here; for portfolio allocation the more natural concept is *concept drift* — the relationship between input features and optimal portfolio weights changes over time and differs across markets.

The financial forecasting context adds a fifth type of heterogeneity not well studied in the generic FL literature: *temporal regime heterogeneity*. Nosedá et al. [9] study FL for financial time series under IID and temporally-partitioned non-IID settings and find that FedAvg significantly outperforms local-only training under the non-IID temporal partition, achieving an 18.8% accuracy improvement over isolated clients. However, they report only aggregate metrics across all clients and do not disaggregate by market size or volatility, which is precisely the analysis our project contributes.

2.3. Personalised FL: Local Adaptation Strategies

Standard FL produces a single global model for all clients. Personalised FL (PFL) instead finds a global initialisation or shared backbone that each client can quickly adapt to its own distribution. The most influential approach is Per-FedAvg, proposed by Fallah et al. [3]. Rather than minimising the average loss across clients, Per-FedAvg optimises for a shared initialisation from which each client can quickly adapt with one or a few gradient steps. In effect, it finds a starting point that is easy to fine-tune, not a point that already works for everyone. Per-FedAvg shows measurable gains over standard FedAvg when client distributions are genuinely different.

A more recent ICLR 2023 paper, FedPAC by Xu et al. [4], extends the personalisation idea by learning both a shared global feature representation and a person-

alised classifier head per client. FedPAC is evaluated against a range of personalised FL baselines including pFedMe, Ditto, FedPer, and FedRep, and shows consistently strong performance under varying levels of label heterogeneity and in data-scarce settings. For our project, the specific insight from FedPAC is that separating the shared backbone (which captures universal market patterns such as momentum and mean-reversion) from the local head (which captures market-specific dynamics) is both technically principled and practically useful.

The survey of Kulkarni et al. [6] provides useful context on the broader PFL landscape. A key practical observation is that simple FedAvg followed by a few local fine-tuning steps on each client’s own data is a widely used and reproducible baseline for personalisation. This is *distinct from* full Per-FedAvg, which requires second-order MAML updates; for an initial semester project, starting with local fine-tuning and comparing it against standard FedAvg is the more tractable choice, with full Per-FedAvg as a potential extension.

2.4. FL Applied to Finance and Portfolio Optimisation

The application of FL to finance has grown rapidly but remains focused on fraud detection and credit risk rather than portfolio optimisation. A dedicated survey on FL in financial security [11] explicitly classifies collaborative portfolio optimisation as a low-exposure FL application where institutions with limited data can leverage global market intelligence without sharing proprietary information.

The most direct prior work on federated portfolio management is Yu et al. [7], which combines FL with reinforcement learning on S&P 500 data, demonstrating that personalised and robust federated RL produces higher Sharpe ratios than non-federated alternatives. However, all assets in that study are drawn from the same market and the same sector. The federation is therefore artificial in terms of structural heterogeneity, as all clients are sub-samples of the same distribution.

The most technically advanced cross-market FL paper is FQL by Cui et al. [8], which proposes a heterogeneity-aware Federated Quantitative Learning

framework evaluated on real data from the US, EU, and Asia-Pacific. FQL introduces a Path Quality-Aware Aggregation Mechanism, a Gradient Clipping and Compression module, and a Heterogeneity-Adaptive Optimiser, reporting an annualised return of 12.72% and a Sharpe ratio of 1.12, compared to FedAvg’s 9.21% return and 0.87 Sharpe. FQL is the closest existing work to our direction. However, its contribution is primarily an engineering one — it improves how updates are aggregated and compressed, but does not examine whether individual markets benefit equally from federation, or whether the performance gains are distributed fairly across clients.

Open Problems (flagged by the field):

- FQL [8] shows that heterogeneity-aware aggregation outperforms FedAvg in cross-market finance, but reports only aggregate metrics. Based on the reviewed papers, per-client fairness under cross-market FL heterogeneity appears to be an unaddressed problem.
- The reviewed finance-FL papers partition clients by institution or region, but do not address temporal regime shifts (bull/bear, crisis/recovery) as a source of heterogeneity. This is a gap synthesised from the general non-IID literature [5] applied to the finance domain.
- The RL-in-finance review [10] flags weak evaluation standards, poor reproducibility, and lack of standardised benchmarks as primary open problems across 167 surveyed papers.
- Whether personalised FL helps under financial non-stationarity remains an open empirical question. Per-FedAvg [3] provides the theoretical motivation, but no reviewed paper has tested this on cross-market portfolio data.
- The non-IID survey [5] notes that spatiotemporal skew remains the least studied heterogeneity type in FL; financial markets are a natural setting where this skew is structurally present.

3. Research Gap and Proposed Direction

The literature establishes two things clearly. First, standard FedAvg degrades under non-IID data be-

cause client drift causes the global model to be biased toward high-volume or dominant clients. Second, cross-market FL for financial strategy is a recognised application area, and recent papers like FQL show that better aggregation algorithms improve aggregate performance. In the papers reviewed here, however, we did not find a study that breaks down FL performance by individual market or client type. FQL and related papers report one pooled Sharpe ratio. They do not ask: does the KSE-100 client gain or lose from participating alongside the S&P 500? Is the benefit of federation roughly equal across markets, or does it flow predominantly to clients that already resemble the majority? And if disadvantaged clients exist, is lightweight personalisation sufficient to correct the imbalance?

We choose this gap over other candidates (such as communication efficiency or privacy-preserving aggregation) because it is (a) directly addressable with public data and standard tools within one semester, (b) scientifically interesting regardless of outcome, and (c) practically important for any real-world deployment of FL in emerging market finance.

Research Question (formal): *When portfolio allocation models are trained via federated learning across markets with structurally heterogeneous return distributions, does standard FedAvg produce asymmetric benefits across clients, and does personalised FL (FedAvg followed by local fine-tuning steps) recover performance for disadvantaged clients, particularly those from smaller or more volatile market environments?*

4. Benchmark, Evaluation, and Data Landscape

Datasets. All price data will be sourced from Yahoo Finance via the `yfinance` Python library, which is free and requires no institutional approval. Four market indices will serve as the federated clients. For each market, a fixed universe of 10–15 large-capitalisation constituent stocks will be selected and held constant throughout the study period. The stock universe will be drawn from publicly documented historical constituent lists where available (e.g., Wikipedia historical S&P 500 changes, LSE FTSE official archives);

where historical lists are unavailable, a stable subset of stocks continuously traded from 2010 will be used. This approach mitigates survivorship bias to the extent practically possible with public data sources. Table 1 summarises the planned datasets.

Table 1: Market datasets used as federated clients.

Dataset	Period	Assets	Notes
S&P 500 (US)	2010–26	15 stocks	Large, liquid, USD
FTSE 100 (UK)	2010–26	15 stocks	Medium, GBP, post-Brexit
KSE-100 (PK)	2010–26	10 stocks	Small, PKR, high vol.
Nikkei 225 (JP)	2010–26	15 stocks	Large, JPY, export-driven

Evaluation Protocol. The field uses a standard set of financial performance metrics. *Annualised return* measures raw profitability over the test period. *Sharpe ratio* is the primary risk-adjusted metric: it divides excess return (above the risk-free rate) by the standard deviation of returns, rewarding consistent performance over lucky outcomes. *Maximum drawdown* measures the worst peak-to-trough loss experienced during the test period and captures tail risk that Sharpe ratio misses. *Portfolio turnover* measures how frequently the model rebalances and serves as a proxy for transaction cost sensitivity. These four metrics together are the standard suite in portfolio optimisation papers and will be computed *separately per client*, not pooled across markets. The train/validation/test split will be strictly chronological, with 2010–2019 for training, 2020–2021 for validation, and 2022–2026 for held-out testing. Transaction costs of 0.1% per rebalance will be applied to all strategies.

Compute Context. The FQL paper [8] used RTX 3090 GPUs for client nodes and a central server with dual A100 GPUs and 512 GB RAM. Our experiments are designed for Google Colab (T4 GPU, 16 GB VRAM), which is sufficient for LSTM or compact Transformer models on daily financial time series of the scale described.

Experimental Design Considerations

Address what you can from the following:

- **Baselines required.** Six baselines are needed to make the results credible: (1) equal-weight portfolio, the trivial benchmark that any learned model must beat; (2) mean-variance optimisation, the classical Markowitz baseline; (3) local-only training, where each client trains independently with no federation; (4) centralised training, where all data is pooled (unrealistic in practice but a useful upper bound); (5) standard FedAvg without personalisation; and (6) FedProx. The key comparison — FedAvg versus personalised FL measured *per client* — is the central contribution of the project.
- **Meaningful ablations.** Three ablations will isolate the key variables. First, vary the number of local fine-tuning steps in personalized FL (0, 1, 3, 5 steps) to understand the sensitivity of the personalisation benefit. Second, vary the composition of the federation: train with only developed markets (US + UK + JP) and compare to the full four-client federation including KSE-100, to isolate the effect of including a heterogeneous emerging market on developed-market performance. Third, compute a heterogeneity score (e.g., pairwise return correlation or distribution divergence) between client datasets and correlate it with the size of the FL benefit gap, to provide a quantitative explanation for any asymmetry observed.
- **Evaluation pitfalls specific to this problem.** Finance ML has well-documented failure modes. *Look-ahead bias* is the most dangerous: any use of future data in feature construction or train/test splits will produce artificially strong results. All splits must be strictly chronological, and all features must be computable using only information available at prediction time. *Survivorship bias* arises if asset universes are selected based on which stocks survived to the end of the sample period; we will use historically documented constituent lists where available, otherwise a fixed stable subset of stocks continuously traded from 2010. *Overfitting to the test period* is common in short backtests; our four-year held-out test period (2022–2026) spans multiple distinct market regimes including post-COVID

recovery, inflation shock, and tech-sector correction, providing a demanding and varied evaluation environment.

5. Annotated Reading List

References

- [1] H. B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y Arcas. Communication-efficient learning of deep networks from decentralized data. *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2017. *What it does: proposes Federated Learning and the FedAvg algorithm, where clients train locally and a server aggregates weight updates by sample-weighted averaging across communication rounds. What it contributes: demonstrates 10–100× reduction in communication rounds compared to naive gradient sharing, with results on MNIST, CIFAR-10, and language modeling. Why it is here: foundational paper defining FL and FedAvg, which is the primary baseline for this project and the starting point for understanding the non-IID problem.*
- [2] T. Li, A. K. Sahu, M. Zaheer, M. Sanjabi, A. Talwalkar, and V. Smith. Federated optimization in heterogeneous networks. *Proceedings of Machine Learning and Systems (MLSys)*, 2020. *What it does: introduces FedProx, which adds a proximal regularization term to each client’s local objective to limit deviation from the current global model during local training. What it contributes: provides convergence guarantees for FL under the combination of statistical and system heterogeneity with partial device participation, and shows empirical improvements over FedAvg on non-IID benchmarks, though gains depend on careful tuning of the proximal coefficient. Why it is here: FedProx is one of the six required baselines in this project’s experimental design and provides the theoretical grounding for why FedAvg underperforms under cross-market heterogeneity.*
- [3] A. Fallah, A. Mokhtari, and A. Ozdaglar. Personalized federated learning: A meta-learning

- approach. *Advances in Neural Information Processing Systems (NeurIPS)*, 2020. What it does: reformulates FL as a meta-learning problem where the global model is an initialisation point that each client can rapidly adapt to its own data with one or a few gradient steps, calling this Per-FedAvg. What it contributes: shows Per-FedAvg achieves better local performance than FedAvg under heterogeneous data distributions, with convergence analysis in the non-convex setting. Why it is here: Per-FedAvg motivates the personalisation direction of this project. The project implements the simpler variant of FedAvg followed by local fine-tuning steps, which is conceptually related to but less expensive than full Per-FedAvg. Full Per-FedAvg with MAML-style second-order updates is a natural FYP extension.
- [4] J. Xu, X. Tong, and S.-L. Huang. Personalized federated learning with feature alignment and classifier collaboration. *International Conference on Learning Representations (ICLR)*, 2023. What it does: decouples the global model into a shared feature backbone and a per-client classification head, jointly optimizing representation alignment across clients and local head adaptation. What it contributes: demonstrates consistently strong performance against personalised FL baselines including pFedMe, Ditto, FedPer, FedRep, and LG-FedAvg across multiple non-IID heterogeneity levels, with particular robustness when per-client data is scarce. Why it is here: provides the architectural motivation for separating a shared temporal backbone (capturing universal market patterns) from a per-client allocation head (capturing market-specific dynamics).
- [5] D. M. Jimenez G., D. Solans, M. Heikkilä, A. Vitaletti, N. Kourtellis, A. Anagnostopoulos, and I. Chatzigiannakis. Non-IID data in federated learning: A survey with taxonomy, metrics, methods, frameworks and future directions. *arXiv preprint arXiv:2411.12377*, 2024. What it does: provides a systematic review of 235 papers on non-IID FL, offering a detailed taxonomy of data heterogeneity types including quantity skew, label distribution skew, feature distribution shift, and spatiotemporal heterogeneity, along with partition protocols and metrics. What it contributes: establishes that quantity imbalance and distribution skew interact in FL, and provides standardised metrics for quantifying non-IIDness, including the Hellinger Distance as a principled measure of client distribution divergence. Why it is here: provides the theoretical framework for classifying the cross-market financial heterogeneity that this project studies, in particular spatiotemporal skew, which closely matches the structural differences between financial market clients.
- [6] V. Kulkarni, M. Kulkarni, and A. Pant. Survey of personalization techniques for federated learning. *2020 Fourth World Conference on Smart Trends in Systems, Security and Sustainability (WorldS4)*, pp. 794–797, 2020. What it does: surveys personalised FL methods including local fine-tuning, meta-learning, multi-task learning, and mixture-of-experts approaches, and discusses why a one-size-fits-all global model fails under statistical heterogeneity. What it contributes: a structured overview of the two-step personalisation framework: first build a global model collaboratively, then adapt it locally using each client’s private data. Why it is here: motivates the use of local fine-tuning as a simple and widely applicable personalisation baseline, and provides conceptual grounding for why global model adaptation is needed when clients have structurally different data distributions.
- [7] P. Yu, L. Wynter, and S. H. Lim. Federated reinforcement learning for portfolio management. In Ludwig and Baracaldo (eds.), *Federated Learning*. Springer, 2022. What it does: applies personalized, robust federated RL to portfolio management using 50 assets from the S&P 500 technology sector, showing higher Sharpe ratios than non-federated alternatives. What it contributes: the first demonstration that FL + RL for portfolio management is feasible and beneficial, using Fed+ for personalization and robust aggregation. Why it is here: the most direct predecessor to this project; its limitation is that all 50 assets come from one market and one sector,

so the federation is not genuinely cross-market.

- [8] W. Cui, L. Zhang, Z. Sun, Z. Zhai, X. Cai, Z. Lan, and Y. Zhan. A federated learning framework with attention mechanism and gradient compression for time-series strategy modeling. *Electronics*, 14(16), 2025. *What it does: proposes FQL, a heterogeneity-aware FL framework for cross-market financial strategy modelling, combining a path-quality-aware aggregation mechanism, gradient clipping and compression, and a heterogeneity-adaptive optimiser, tested on data from the US, EU, and Asia-Pacific regions. What it contributes: achieves 12.72% annualised return and a Sharpe ratio of 1.12, compared to FedAvg’s 9.21% return and 0.87 Sharpe, demonstrating that heterogeneity-aware aggregation outperforms uniform averaging on cross-market financial data. Why it is here: the closest existing work to this project; its contribution is an engineering improvement to aggregation and compression, but it reports only aggregate metrics without per-client fairness analysis, which is the specific gap this project addresses.*
- [9] M. Nosedá, A. De Luca, L. Von Briel, and N. Lacour. Federated learning for financial forecasting. *arXiv preprint arXiv:2509.16393*, 2025. *What it does: studies FL for financial market prediction under both IID and temporally-partitioned non-IID data on S&P 500, showing that FedAvg achieves an 18.8% accuracy improvement over local-only models under the non-IID temporal partition setting. What it contributes: demonstrates that temporal heterogeneity amplifies the benefit of federation relative to isolated training, but only reports aggregate metrics across all clients rather than disaggregating performance by client. Why it is here: empirically motivates the project by showing that the non-IID structure of financial data matters for federation benefit, while leaving open the question of which specific clients gain most and whether any are disadvantaged.*
- [10] M. R. Hoque, M. M. Ferdous, and M. K. Hasan. Reinforcement learning in financial decision making: A systematic review of performance, challenges, and implementation strategies. *arXiv preprint arXiv:2512.10913*, 2025. *What it does: reviews 167 papers from 2017–2025 on RL in finance, covering portfolio optimization, market making, and algorithmic trading, assessing implementation quality and evaluation standards. What it contributes: concludes that implementation quality and domain knowledge outweigh algorithmic sophistication, and that the field needs standardized benchmarks, robustness testing, and per-environment evaluation. Why it is here: calibrates the project’s evaluation ambition; the per-client analysis and regime-split evaluation design in this project directly respond to the benchmarking gaps this review identifies.*
- [11] C. H. Kennedy, A. Hilal, and M. Momeni. The role of federated learning in improving financial security: A survey. *IEEE Global Conference on Artificial Intelligence and Internet of Things (GCAIoT)*, 2025. Also available as *arXiv:2510.14991*. *What it does: surveys FL applications in finance, classifying them by regulatory exposure from low-risk internal use cases such as collaborative portfolio optimisation to high-risk applications such as anti-money laundering detection. What it contributes: explicitly identifies collaborative portfolio optimisation as a low-exposure FL application where institutions can co-train on local datasets while preserving confidentiality of proprietary trading data, client profiles, and risk metrics. Why it is here: provides the primary citation for categorising portfolio optimisation as a recognised FL application domain and motivating the institutional privacy argument at the core of this project.*

6. Preliminary Implementation Resources

- <https://github.com/adap/flower>
Flower (flwr): the FL simulation framework referenced in Section 2.5 of the course project document. Actively maintained, handles the client-server communication loop and aggregation strategies including FedAvg and FedProx out of the box. Will be used as the primary FL infrastructure for this project.

- <https://github.com/AI4Finance-Foundation/FinRL> FinRL: a widely-cited deep reinforcement learning framework for financial trading. Contains data pipelines for Yahoo Finance and portfolio environment abstractions. Relevant as a reference for the local-only and centralised training baselines; it is not a federated learning framework and will not be used for the FL experiments themselves.
- <https://github.com/google-research/federated> A Google Research collection of projects related to federated learning and federated analytics. Examined for reference implementations of FL concepts. Note: this is not the TensorFlow Federated (TFF) framework itself but a collection of research prototypes; descriptions of specific built-in algorithms should be verified against individual project READMEs before use.
- <https://github.com/lx10077/fedavgpy> Lightweight PyTorch reimplementation of FedAvg with non-IID data partitioning utilities. Useful for understanding the FedAvg training loop at the code level and for testing non-IID partition strategies before integrating into the Flower framework.
- <https://github.com/PaddlePaddle/PaddleFL> PaddleFL: a federated learning framework supporting FedAvg and differential-privacy-based training. Examined for reference on federated training loop structure. Specific personalised FL variants should be verified directly in the codebase before relying on them.
- <https://github.com/microsoft/qlib> Qlib: Microsoft’s quantitative investment research platform. Provides data loading, backtesting infrastructure, and portfolio strategy utilities. Useful as a reference for implementing equal-weight and mean-variance baselines; market-specific data availability should be verified directly in the platform’s data documentation.
- <https://github.com/mtuann/federated-learning-updated-papers> A curated and actively updated repository of FL

papers from top venues. Used during the literature survey to identify recent work on personalised FL, non-IID heterogeneity, and FL for financial applications. Not a code resource but a useful secondary discovery tool for related work tracking.